

Java Interface & Inheritance Programs

Complete code with corresponding terminal-style output

1. Single Inheritance — Animal and Dog

CODE

```
package interfaces;
// Parent class
class Animal
{
    // Method of parent class
    void eat()
    {
        System.out.println("Animal eats");
    }
}

// Child class extends Parent class
class Dog extends Animal
{
    // Method of child class
    void bark()
    {
        System.out.println("Dog barks");
    }
}

// Main class
class InheritanceDemo
{
    public static void main(String args[])
    {
        // Creating object of Dog class
        Dog d = new Dog();

        // Calling inherited method from Animal
        d.eat();

        // Calling Dog's own method
        d.bark();
    }
}
```

OUTPUT

```
C:\> java Program
```

```
Animal eats
Dog barks
```

2. Invalid Multiple Inheritance Using Classes

CODE

```
package interfaces;
class C2{
    void show2(){
        System.out.println("C2 method");
    }
}
class C3{
    void show3(){
        System.out.println("C3 method");
    }
}
// ERROR: Java does not allow extending two classes
class C1 extends C2, C3{
    void show1(){
        System.out.println("C1");
    }
}
```

OUTPUT

```
C:\> java Program
```

COMPILATION ERROR

```
Java does not allow a class to extend two classes.
Invalid syntax: class C1 extends C2, C3
```

3. Class Inheritance with Interface Implementation

CODE

```
package interfaces;
// Parent class
class C2
{
    // Method of C2 class
    void show2()
    {
        System.out.println("C2 method");
    }
}

// Interface
interface I1
{
    // Abstract method of I1
    void show1();
}

// C1 inherits C2 and implements I1
class C1 extends C2 implements I1
{
    // Implementing show1() method of I1
    public void show1()
    {
        System.out.println("I1 method");
    }

    // Main method
    public static void main(String args[])
    {
        // Creating object of C1 class
        C1 obj = new C1();

        // Calling show2() inherited from C2
        obj.show2();

        // Calling show1() implemented from I1
        obj.show1();
    }
}
```

OUTPUT

```
C:\> java Program
```

```
C2 method
```

```
I1 method
```

4. Class Inheritance with Multiple Interfaces

CODE

```
package interfaces;
// Parent class
class C2
{
    // Method of C2 class
    void show2()
    {
        System.out.println("C2 method");
    }
}

// First interface
interface I1
{
    // Abstract method of I1
    void show1();
}

// Second interface
interface I2
{
    // Abstract method of I2
    void showI2();
}

// Third interface
interface I3
{
    // Abstract method of I3
    void showI3();
}

// C1 inherits C2
// and implements three interfaces: I1, I2 and I3
class C1 extends C2 implements I1, I2, I3
{
    // Implementing show1() method of I1
    public void show1()
    {
        System.out.println("I1 method");
    }

    // Implementing showI2() method of I2
    public void showI2()
    {
        System.out.println("I2 method");
    }

    // Implementing showI3() method of I3
    public void showI3()
    {
        System.out.println("I3 method");
    }

    // Main method
    public static void main(String args[])
    {
        // Creating object of C1 class
        C1 obj = new C1();

        // Calling show2() inherited from C2
        obj.show2();

        // Calling show1() implemented from I1
        obj.show1();

        // Calling showI2() implemented from I2
        obj.showI2();

        // Calling showI3() implemented from I3
        obj.showI3();
    }
}
```

OUTPUT

```
C:\> java Program
```

```
C2 method
```

```
I1 method
```

```
I2 method
```

```
I3 method
```

5. Invalid Use of extends and implements

CODE

```
package interfaces;
class C2{
    void show2(){
        System.out.println("C2 method");
    }
}
interface I1{
    void show1();
}
// Invalid statement
class C1 implements C2 extends I1{
    public void show1(){
        System.out.println("I1 method");
    }
}
public class Main{
    public static void main(String args[]){
        C1 obj = new C1();
        obj.show2();
        obj.show1();
    }
}
```

OUTPUT

```
C:\> java Program
```

```
COMPILATION ERROR
```

```
C2 is a class, so C1 must use extends C2.
```

```
I1 is an interface, so C1 must use implements I1.
```

```
Correct syntax: class C1 extends C2 implements I1
```

6. Basic Interface Implementation

CODE

```
package interfaces;
interface I1{
    void show1();
}
class C1 implements I1{
    public void show1(){
        System.out.println("I1 method");
    }
}
public class Main{
    public static void main(String args[]){
        C1 obj = new C1();
        obj.show1();
    }
}
```

OUTPUT

```
C:\> java Program
```

```
I1 method
```

7. Class Inheritance with Two Interfaces — book and card

CODE

```
package interfaces;
interface book{
    int x=10;
    void name();
}
interface card{
    int z=15;
    void num();
}
class ram{
    void show() {
        System.out.println("shoe the data");
    }
}
class raj extends ram implements book,card{
    @Override
    public void name() {
        System.out.println("i am siva");
    }
    public void num() {
        System.out.println("num 000");
    }
}
}
public class Sample {
    public static void main(String[] args) {
        raj r = new raj();
        r.show();
        r.num();
        r.name();
    }
}
```

OUTPUT

```
C:\> java Program

shoe the data
num 000
i am siva
```


8. Interface Inheritance — raji extends riya

CODE

```
package interfaces;
interface riya{
    int x=10;
    void siva();
}
interface raji extends riya {
    int z=15;
    void lucky();
}
class Raghu implements raji{
    @Override
    public void siva() {
        System.out.println("i am siva");
    }
    @Override
    public void lucky() {
        System.out.println("i am lucky");
    }
}
public class Simple {
    public static void main(String[] args) {
        Raghu r=new Raghu();
        r.siva();
        r.lucky();
    }
}
```

OUTPUT

```
C:\> java Program
```

```
i am siva
i am lucky
```

9. Multiple Interface Inheritance — I3 extends I1 and I2

CODE

```
package interfaces;
interface I1{
    void ex();
}
interface I2{
    void in();
}
interface I3 extends I1, I2{
    void lee();
}
class Raw implements I3{
    @Override
    public void ex() {
        System.out.println("this is from I1");
    }
    @Override
    public void in() {
        System.out.println("this is from I2");
    }
    @Override
    public void lee() {
        System.out.println("this is from I3");
    }
}
public class Simple1 {
    public static void main(String[] args) {
        Raw a = new Raw();
        a.ex();
        a.in();
        a.lee();
    }
}
```

OUTPUT

```
C:\> java Program

this is from I1
this is from I2
this is from I3
```